

William Wong

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EDUCATION

University of Southern California
Master of Science, Electrical Engineering.

Los Angeles, CA
May 2026

University of Southern California
Bachelor of Science, Electrical and Computer Engineering.

Los Angeles, CA
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EXPERIENCE

Electrical Design Intern

May 2025 – Aug 2025

Yu Ping Enterprise

Kaohsiung, TW

- Modeled switchgear lineups, panelboards, and conduit routing for one floor in Revit, verified NEC clearances, and used Navisworks to resolve 30 HVAC and plumbing conflicts before installation.
- Developed LV relay control circuits and BAS interface points for 2 AHUs and 24 VAVs, producing wiring diagrams and I/O point lists for mechanical and controls lead markup.
- Built Excel VBA macros to parse commissioning logs, flag voltage deviations above 5%, and tabulate phase imbalance by zone, streamlining a 6-hour weekly review.

Hardware Engineering Intern

May 2024 – Aug 2024

Droxo Tech

Tainan, TW

- Designed a spray-flow sensor interface board in KiCad, integrating a Hall flowmeter input with a 4.7 k Ω pull-up, 1 kHz RC filter, INA219 monitor on a 0.033 Ω shunt, and a 24 V TVS at the harness connector.
- Characterized a PWM-controlled pump across three nozzle configurations, generating duty-cycle-to-flow calibration curves from 0.4–2.1 L/min with $\pm 5\%$ flow accuracy.
- Investigated 25 returned motor driver boards, tracing failures to electrochemical migration, MOSFET overheating, and connector fretting; measured a 118 $^{\circ}\text{C}$ hotspot at 8 A that justified a copper-pour increase.

Electrical Engineering Intern

May 2023 – Aug 2023

Skymul

Peachtree Corners, GA

- Designed DRV8353 gate-drive and power-input blocks in Altium, selecting 10 Ω gate resistors, 1 μF VCP, and 47 nF bypass caps.
- Simulated a 24 V input transient in LTspice to size bulk capacitance for a 5 A load step, then routed high-current motor phases and a 120 Ω CAN pair on a four-layer PCB.
- Ran a 40-point validation matrix across 0–6 A locked-rotor limits, 20–26 V input, and 0–55 $^{\circ}\text{C}$ to qualify actuator-driver builds.

PROJECTS

USC Formula SAE, Electronics Team

- Designed the tractive-system LV harness, terminating 120 sealed contacts with twisted-pair CAN routing and isolated returns.
- Validated the shutdown circuit across 50 simulated BSPD, BOTS, IMD, and E-Stop faults, confirming AIR opening under 10 ms and IMD trip at 500 Ω/V .
- Sized the precharge resistor for a 5-time-constant rise into the 400 μF DC-link capacitance, limiting inrush to 8 A and confirming 95% bus voltage before AIR closure.
- Systematized 100 Hz LabVIEW and CAN logging of pack voltage, phase current, and brake pressure.

CPRPal

- Designed a CPR feedback device on a custom PCB fusing an HX711 load cell, BNO055 IMU, and PPG sensor.
- Built adaptive-baseline PPG detection with hysteresis and contact validation to reject compression artifact across both ESP32-S3 cores under FreeRTOS.

Air Quality Monitor

- Built a desktop AQI monitor driving a servo petal linkage that blooms in clean air and withers as quality degrades.
- Integrated a ENS161 gas sensor with an ENS210 temp/humidity sensor on a shared I2C bus for on-chip compensation, with threshold hysteresis to stop linkage oscillation.

8-Channel Peristaltic Pump Controller

- Designed a 2-layer board driving eight 12 V pumps through four TB6612FNG H-bridges, sizing low-side MOSFET switching and decoupling to hold the rail within 200 mV under inductive inrush.
- Drafted a HUSB238 USB-PD sink stage with back-to-back MOSFET load switching to negotiate 5, 9, 12, and 20 V.

SKILLS

EDA and PCB Design: Altium, KiCad, LTspice, ANSYS

Communication Protocols: SPI, I2C, CAN

CAD and Design Tools: AutoCAD, Revit, Navisworks

Programming and Scripting: Python, C++, MATLAB, Excel VBA